Mohammad Yazdani

Tehran, Iran | mohamad.yazdani7990@gmail.com | LinkedIn | Website | +98 902 816 4610

EDUCATION

UNIVERSITY OF TEHRAN Tehran, Iran
M.Sc., Process Engineering — GPA: 3.63/4 *Sep 2023 – Present*
Thesis: *Surrogate-Assisted Multi-Objective Optimization of an Integrated Adsorption-Cryogenic CO2 Capture Process for Steam Methane Reforming Flue Gas*

IRAN UNIVERSITY OF SCIENCE AND TECHNOLOGY Tehran, Iran
B.Sc., Chemical Engineering — GPA: 3.64/4 *Sep 2019 – Sep 2023*
Thesis: *From Model to Market: A Feasibility Study of Acrylic Ester Production*

RESEARCH INTERESTS

• Process Systems Engineering (PSE) • Process Modeling, Simulation & Digital Twins • Mathematical Optimization • Carbon Capture, Utilization & Storage (CCUS) • Techno-Economic & Sustainability Analysis • Artificial Intelligence for Chemical Engineering

PUBLICATIONS

JOURNAL ARTICLE

- **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. "Techno-Economic Assessment of an Optimized Integrated Adsorption-Cryogenic CO2 Capture Process for Steam-Methane Reforming Flue Gas". **Under review** at the *International Journal of Greenhouse Gas Control*
- **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. "Surrogate-Assisted Multi-Objective Optimization of an Integrated Adsorption-Cryogenic CO2 Capture Process for Steam Methane Reforming Flue Gas". **Under review** at *Journal of CO2 Utilization*

CONFERENCE ARTICLES

- **Yazdani, M.**, Charoughchi, A., Fatemi, S., and Fakhroleslam, M. "Surrogate-Assisted Multi-Objective Optimization of VPSA for CO2 Capture from Steam Methane Reforming Flue Gas." The 13th International Chemical Engineering Congress & Exhibition (IChEC 2026), Tehran, Iran, 2026.
- Charoughchi, A., **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. "Integrated Waste Heat Recovery and Adsorption System for SO2 Removal from Flue Gas." The 13th International Chemical Engineering Congress & Exhibition (IChEC 2026), Tehran, Iran, 2026.
- **Yazdani, M.**, Charoughchi, A., Fatemi, S., and Fakhroleslam, M. "Simulation of a Hybrid Vacuum Swing Adsorption–Cryogenic Process for CO2 Removal from Flue Gas." Iranian Conference on Compressed Gases (IGCC), Tehran, Iran, 2025.
- Charoughchi, A., and **Yazdani, M.** "Simulation of Sulfur Dioxide (SO2) Pollutant Adsorption from High-Sulfur Fuel Combustion." Iranian Conference on Compressed Gases (IGCC), Tehran, Iran, 2025.
- **Yazdani, M.**, Charoughchi, A., Fatemi, S., and Fakhroleslam, M. "Development of Machine Learning-Based Surrogate Models for the Optimization of Direct Air Capture (DAC) Process." 6th International Conference on Soft Computing (ICSC), Tehran, Iran, 2025.
- Charoughchi, A., **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. "Comparative Study of Response Surface Methodology (RSM) and Random Forest (RF) Models for Predicting SO2 Recovery from Flue Gas in an Adsorption Process." 6th International Conference on Soft Computing (ICSC), Tehran, Iran, 2025.

RESEARCH EXPERIENCE

UNIVERSITY OF TEHRAN Tehran, Iran
Research Assistant *Jun 2024 - Present*

- Developed automated workflows for process modeling, simulation, optimization, and digital process engineering using Aspen Adsorption, Aspen Plus, Python, and machine learning.
- Implemented mathematical optimization and AI-enabled surrogate modeling for process design and techno-economic assessment.
- Reduced computational time for process optimization by replacing high-fidelity simulations with surrogate models, enabling rapid exploration of hundreds of operating conditions.

IRAN UNIVERSITY OF SCIENCE AND TECHNOLOGY Tehran, Iran
Research Assistant *Mar 2023 - Sep 2023*

- Led an independent techno-economic and market analysis of five competing industrial production technologies for acrylate esters (Dow, BASF, Toyo Soda, Distillers, Arkema), synthesizing process data, economic modeling, and patent life-cycle analysis into a unified evaluation framework.
- Designed and applied a multi-criteria decision-making methodology (AHP) to systematically compare licensing technologies across economic, market, and technical criteria, resulting in a rigorously justified technology selection.
- Developed a detailed process simulation in Aspen Plus and completed equipment design for the selected process, demonstrating the ability to translate technical literature into a validated, working process model.

TEACHING EXPERIENCE

UNIVERSITY OF TEHRAN Tehran, Iran
Teaching Assistant – Advanced Engineering Mathematics *Oct 2024 - Feb 2025*

- Assisted in delivering tutorials and mentoring students in advanced engineering mathematics, guiding analytical and numerical solution methods.
- Developed and delivered hands-on MATLAB workshops covering numerical methods, mathematical modeling, and engineering problem solving.

UNIVERSITY OF TEHRAN Tehran, Iran
Teaching Assistant – Advanced Adsorption Processes *Oct 2025 - Feb 2026*

- Assisted in teaching advanced adsorption processes by mentoring students in adsorption theory, process modeling, and simulation concepts.
- Developed and delivered hands-on Aspen Adsorption workshops covering process model development, cycle simulation, and process performance evaluation.

PROFESSIONAL EXPERIENCE

PETRO RAMAN TECHNOLOGY (PRT) CO. Tehran, Iran
Process Engineer *Jun 2023 - Present*

- Led the Process Engineering discipline for feasibility, conceptual design, basic and detailed engineering, and debottlenecking studies across petrochemical projects.
- Performed process simulation, engineering calculations, equipment specification, and engineering document review.
- Collaborated with multidisciplinary engineering teams and licensors throughout project execution.

SHAZAND PETROCHEMICAL COMPANY (AR.P.C) Shazand, Iran
Internship *Jul 2022 - Sep 2022*

- Gained hands-on process engineering experience in an operating Olefin unit, working from Technip licensor design documents to perform equipment sizing, hydraulic calculations, and process simulation (Aspen HYSYS/Plus), with direct exposure to live plant operations.

TECHNICAL SKILLS

Programming: Python, MATLAB, FORTRAN

Process Simulation & Engineering Software: Aspen Plus, Aspen HYSYS, AVEVA Process Simulation, DWSIM, UniSim, Aspen Custom Modeler, Aspen Adsorption, Aspen Process Economic Analyzer, Aspen Flare System Analyzer, Aspen Exchanger Design & Rating, HTRI, PIPEFLO, AFT Impulse & Fathom, CONVAL, COMSOL Multiphysics

General: AutoCAD, Microsoft Office, Adobe Products, Figma

Languages: Persian (native), English (**C1** – TOEFL iBT **100**: R**27**, L**25**, S**24**, W**24**)

LICENSES & CERTIFICATIONS

AI FOR EVERYONE, BY ANDREW NG *Nov 2023*

- Issued by DeepLearning.AI (Show credential)

AI PYTHON FOR BEGINNERS, BY ANDREW NG *Oct 2024*

- Issued by DeepLearning.AI

MACHINE LEARNING SPECIALIZATION, BY ANDREW NG *Nov 2024 - Feb 2025*

- Issued by DeepLearning.AI, Stanford University (Show credential)

REFERENCES

DR. SHOHREH FATEMI M.Sc. Supervisor

- Professor at Department of Polymer and Chemical Engineering, University of Tehran, Tehran, Iran, shfatemi@ut.ac.ir

DR. MOHAMMAD FAKHROLESLAM M.Sc. Co-Supervisor

- Assistant Professor at Tarbiat Modares University, Tehran, Iran, fakhroleslam@modares.ac.ir

DR. NOROLLAH KASIRI B.Sc. Supervisor

- Professor at Iran University of Science and Technology, Tehran, Iran, kasiri@iust.ac.ir